DocuMind AI — Multi-Page Test Document

A practice document for testing PDF summaries, document-grounded questions, page references, and fallback behavior.

How to use this PDF

Upload this document to DocuMind AI. Try asking questions whose answers appear on different pages, then ask one question that the document does not answer. The facts below are deliberately distributed across sections to test retrieval.

Contents

- Page 2: Python fundamentals and common uses
- Page 3: SQL and relational databases
- Page 4: Artificial intelligence and machine learning
- Page 5: A fictional bookstore dataset and analysis task
- Page 6: Document search, evaluation, and limitations
- Page 7: Review questions and project requirements

1. Python Fundamentals

Python is a general-purpose programming language known for readable syntax. It is used in automation, web development, data analysis, scientific computing, and artificial intelligence. Python code is commonly organized into variables, functions, loops, conditional statements, and modules.

Important concepts

- **Variable:** a name that refers to a value, such as `total = 25`.
- **Conditional:** an `if` statement selects which code runs based on a condition.
- **Loop:** a `for` or `while` loop repeats instructions.
- **Function:** a reusable block of code defined with `def`.
- **List:** an ordered collection that can hold multiple items.

Example

A program can calculate the total price of several items by adding their prices in a loop. A function named `calculate_total` could accept a list of prices and return their sum.

Python for data work

Libraries extend Python's capabilities. Pandas is commonly used for tabular data, Matplotlib for plotting, and scikit-learn for traditional machine-learning workflows. A library is an additional collection of code that a program can import.

2. SQL and Relational Databases

SQL stands for Structured Query Language. It is used to store, retrieve, and manage data in relational databases. A relational database organizes information into tables containing rows and columns.

Core SQL operations

- **SELECT:** retrieves data from one or more tables.
- **INSERT:** adds new rows.
- **UPDATE:** changes values in existing rows.
- **DELETE:** removes rows.
- **CREATE TABLE:** defines a table and its columns.

Example table: Books

Imagine a table named `Books` with columns `book_id`, `title`, `category`, and `price`. A query such as `SELECT title, price FROM Books;` returns the title and price columns for all rows.

Filtering and grouping

A WHERE clause filters rows according to a condition. ORDER BY sorts results. GROUP BY combines rows with matching values so aggregate functions such as COUNT, SUM, and AVG can summarize them.

A primary key identifies a row uniquely. A foreign key links a value in one table to a key in another table. These constraints help maintain data consistency.

3. Artificial Intelligence and Machine Learning

Artificial intelligence (AI) is a broad field concerned with systems that perform tasks associated with abilities such as understanding language, recognizing patterns, planning, or making predictions. Machine learning (ML) is a subset of AI in which a model learns patterns from data.

Typical machine-learning workflow

- Define the problem and choose a measurable objective.
- Collect relevant data and inspect its quality.
- Prepare features and split data into training and evaluation sets.
- Train a model using the training data.
- Evaluate the model on data not used for training.
- Monitor performance and update the system when needed.

Generative AI

Generative AI models create new content such as text, code, images, or summaries. A large language model can answer questions in natural language. Its output can still contain errors, so applications should ground responses in reliable context and avoid presenting unsupported claims as facts.

Retrieval-augmented generation (RAG)

A RAG application retrieves relevant passages from a source and supplies them to a language model as context. In a PDF assistant, the system can extract text, split it into chunks, retrieve chunks related to a question, and ask the model to answer using those chunks. Page labels can help users trace claims back to the document.

4. Fictional Bookstore Dataset

The following small dataset is fictional and exists only for practice. The bookstore is called Maple Leaf Books. It sells books online and records each order in a table named `Orders`.

Example records

Order ID	Category	Quantity	Unit Price (■)	City
1001	Python	2	450	Hyderabad
1002	SQL	1	500	Warangal
1003	AI	3	650	Hyderabad
1004	Python	1	450	Karimnagar
1005	SQL	2	500	Hyderabad

Business questions

- Which city has the most orders in this sample?
- What is the revenue for order 1003 before any discounts or costs?
- Which categories appear in the records?

Revenue is calculated as quantity multiplied by unit price. Revenue is not the same as profit: calculating profit requires cost information and any relevant expenses.

5. Document Search and Evaluation

A PDF question-answering system should answer from the uploaded document rather than silently substituting outside knowledge. When a question is not supported by the document, the system should clearly state that the information is not available in the source.

Retrieval quality

Chunking divides long documents into smaller passages. Overlap can preserve context that crosses chunk boundaries. Embeddings represent text as numerical vectors, and similarity search can identify passages whose meaning is close to a question.

Useful evaluation checks

- **Grounded answer:** does the response match the document?
- **Page reference:** can a reader locate the supporting passage?
- **Missing-information behavior:** does the system admit when the answer is absent?
- **Fallback behavior:** if the primary model fails, does the app try a backup model before local search?
- **Consistency:** does the same question produce a clear answer across repeated tests?

Known limitations

Scanned PDFs may contain images rather than extractable text. Tables can be difficult to interpret if their layout is not preserved during extraction. Long documents may need careful chunking to fit model context limits. Similarity retrieval can return related but insufficient passages, so the answer prompt should require the model to acknowledge missing evidence.

6. Review Questions and Project Requirements

Use these questions to test DocuMind AI. Some answers are in earlier sections of this document; the last question is intentionally not answered here.

Questions

- What does SQL stand for?
- Which Python library is commonly used for tabular data in this document?
- What is the difference between revenue and profit?
- What is the purpose of retrieval-augmented generation?
- What does a primary key do in a relational database?
- What is the fictional bookstore's name?
- Which city has the most orders in the sample dataset?
- Who won the 2024 Olympic gold medal in women's 100-metre sprint? (This document does not provide the answer.)

Suggested test sequence

- Generate a short, medium, and long summary.
- Ask one question about Python, one about SQL, and one about RAG.
- Ask a question about the bookstore table and check whether the answer uses the provided figures.
- Ask the intentionally unsupported question and check that the app admits the document lacks the answer.

End of test document

This document is a synthetic practice resource created to evaluate PDF text extraction and document-grounded answering. The bookstore dataset is fictional.